

Project Design Phase
Problem – Solution Fit Template

Date	19 July 2026
Team ID	
Project Name	
Maximum Marks	Nightjar & Co. – MERN BookStore

Problem–Solution Fit

Description

The Problem–Solution Fit Canvas is used to identify the major challenges faced by customers when purchasing books online and to evaluate how effectively the proposed solution addresses those challenges. It helps ensure that the application's features are designed around real user needs rather than assumptions, enabling the development of a solution that delivers meaningful value.

For **Nightjar & Co. – MERN BookStore**, the canvas focuses on understanding the common pain points experienced by students, readers, and book enthusiasts, such as difficulty in discovering books, complex navigation, lengthy checkout processes, and concerns about secure user authentication. By analyzing these problems, the team designed a user-centric web application that provides advanced search and category filters, an organized book catalog, secure JWT-based authentication, shopping cart functionality, and a streamlined checkout process.

This analysis confirms that the proposed solution aligns with customer expectations by making book discovery, browsing, and purchasing more convenient, efficient, and secure. As a result, the application improves the overall user experience, increases customer satisfaction, and provides an accessible online platform for readers to explore and purchase books anytime and from anywhere.

Problem–Solution Fit Table

Customer Problem	Impact on User	Solution in Nightjar & Co.	Expected Benefit
Difficulty finding desired books	Wastes time searching across multiple platforms	Search by title, author, and category	Faster and easier book discovery
Complicated online purchasing process	Users abandon purchases	Simple cart and checkout workflow	Smooth buying experience
Lack of secure user authentication	Users hesitate to register	JWT-based authentication	Secure login and user confidence
Poor organization of book collections	Difficult to browse books	Categorized book catalog with filtering	Better browsing experience
Limited access to physical bookstores	Books may not be easily available	Web-based bookstore accessible anytime	Convenient access from anywhere

CUSTOMER PROBLEMS

- ❗ Difficult book search
- ❗ Complex checkout
- ❗ Poor navigation
- ❗ Security concerns



Nightjar & Co.
MERN BookStore

- ✅ Search & Filters
- ✅ Secure Login
- ✅ Shopping Cart
- ✅ Organized Categories



USER BENEFITS

- 😊 Saves Time
- 😊 Easy Shopping
- 😊 Secure Experience
- 😊 Better User Satisfaction

≡ PROBLEM-SOLUTION CANVAS ≡

! PROBLEMS	✓ SOLUTIONS
 Hard to find books	 Advanced search and category filters
 Confusing websites	 Clean and intuitive user interface
 Insecure login	 JWT authentication
 Long buying process	 Streamlined cart and checkout
 Limited accessibility	 Responsive web application available anytime